

Evaluation Dataset of Multiple-Choice Cloze Items for Vocabulary Training and Testing

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Figure 1: Multiple-choice cloze item for vocabulary learning

ABSTRACT

Vocabulary learning is a typical part of nearly any second language learning curriculum. This entails methodologies and materials for training and testing vocabulary knowledge in learners. In large-scale programs, the preparation of such materials can be labor intensive and thus automatic means of generation are desirable. VocaTT (Vocabulary Training and Testing) is an ongoing project to use machine learning methods to generate novel multiple choice

cloze (i.e., fill-in-the-blank) items for use in second language learning programs. This paper describes the ongoing creation of a gold standard set of multiple-choice cloze items to be used in training a machine learning algorithm. Machine-generated multiple choice cloze items were reviewed by two experienced language teachers, who evaluated each item for well-formedness (i.e., suitability as multiple-choice cloze test item) with three options: reject as unsalvageable, keep as-is, or revise into a well-formed item as they thought best. Results for a 600-item set that both checkers evaluated show moderate agreement on the question of rejection but slight agreement for keeping as-is. For revised items, the agreement on what type of revisions to make was slight to fair. In an expanded set of 2,792 items, checkers judged most items as needing revision but made varying kinds of revisions to yield well-formed items. Interested researchers may contact the authors to inquire about how they may access and use the evaluation dataset.

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CCS CONCEPTS

• Human-centered computing;

KEYWORDS

rater agreement dataset, multiple-choice cloze question, vocabulary learning, item evaluation

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1 INTRODUCTION

A typical feature of most language learning curricula is a plan for the development of learners' vocabulary knowledge and testing thereof. In this context, multiple choice cloze (i.e., fill-in-the-blank; hereafter, MCC) questions as in (1) are widely used and known to be relatively objective and reliable for testing (cf., [9]).

- (1) Two ancient tombs in this _____ are maintained by the Imperial Household Agency.
a. area b. legality c. benefit
d. principle

The creation of such items is not difficult on a small scale (e.g., by individual teachers for one or two classes at a time) but can be quite labor-intensive as well as inconsistent in large-scale programs where classes are taught by many different teachers in different time slots. The advent of online teaching brought on by the Coronavirus situation adds other difficulties: Once used online, quizzes may be regarded as no longer "secure" and therefore are not reusable with other populations and real-time testing may be insecure as students can potentially share answers through other communication channels. This can be partly solved with large question banks from which each student gets a unique set of random questions, though the creation of such banks is also labor-intensive.

To address these difficulties, we have been engaged in a project to generate MCC questions for vocabulary training and testing called "VocaTT" which involves first, the creation of a gold standard set of MCC questions which will then, second, be used to train a machine learning algorithm which can then, third, produce novel MCC questions. These questions can then be produced on-demand and then adapted immediately for use in online (or offline) training and testing scenarios.

In the present work, we report on the status of the gold standard set by explaining how it has been produced thus far, its current make-up, the status of its evaluation by checkers, and its potential availability to other researchers. The following background section explains some elements that underlie the project. This is followed by a detailed description of the item dataset. The paper concludes with future plans for the dataset and the VocaTT project.

2 BACKGROUND

2.1 Multiple-choice cloze questions

Multiple-choice cloze (MCC) questions as in (1) above and (2)-(3) below are a well-established and widely used question format for testing learner's knowledge in a wide variety of domains (cf., [9]). Each question consists of a short text ('stem') in which some target portion of the text ('key') has been replaced with a blank space. The key (a, c, and b in (1), (2), and (3), respectively) is listed together with several competing options ('distractors') which are sub-optimal completions of the text relative to the key. Although the stem, key, and distractors may be of any length (i.e., a stem of one or more sentences; key and distractors of one or more words), the present work focuses only on MCC questions with one-sentence stems and one-word keys and distractors. Hence, hereafter in this paper, "MCC" refers to this more specific (though still commonly used) case.

- (2) Distortion is _____ area worth consideration.
a. both b. ours c. another
d. all
- (3) There were no other _____ signs of mineral deficiencies.
a. parted b. outward c. below
d. neither

2.2 Automatic generation of MCC questions

There are several proposed and existing systems that can automatically generate MCC from texts (e.g., [8, 11, 15–17, 22]) or from word lists (e.g., [1, 2, 5, 10, 13, 14]). One such system is Word Quiz Constructor (WQC: [18, 19]). Although it is only a pre-cursor to the VocaTT project described in this paper, it served a role in the generation of the gold standard item set and is described here briefly (though interested readers should see [19] for details).

WQC generates a vocabulary quiz with a customizable number of MCC questions, designed to test learners' knowledge of a given list of target words (also customizable by the WQC user). WQC constructs an item by taking a target word from the specified list, drawing a sentence containing the word from an online or offline corpus, and then finding distractor words from the same list. In this process, stems are filtered such that the trigram (i.e., three-word sequence) in the stem with the key word in the center (e.g., "is another area" in (2) and "other outward signs" in (3)) is a high-frequency trigram, while distractors are filtered such that the frequency of the corresponding trigram formed by substituting the key word with a candidate distractor (e.g., "is both area" in (2) and "other parted signs" in (3)) is relatively low. Stems are also filtered by readability scoring in order to control the difficulty of the stem sentences. WQC produces items that are judged well-formed at a rate comparable to items produced manually by an experienced language teacher [19]. Yet, a quiz produced by WQC still needs human review to revise some items that might not be suitable to a particular group of students, or to reject some items that may not be well-formed (e.g., fragmented stem, multiple correct answers; discussed in more detail in section 3.2).

2.3 Word lists

As noted earlier, most language teaching curricula specify some plan for training and testing of vocabulary knowledge. A typical approach is to specify word lists that learners should study over time and then be tested on. These word lists could be of a general nature for broad educational purposes, or specific to a particular domain such as those for use in tourism or medical training. Two widely used lists are described in the subsections further below. A typical way to organize such lists is into word "families", where each family consists of a headword (e.g., *analyze*) along with member words which are directly derived from the headword (e.g., *analysis*, *analytical*, *unanalyzed*).

2.3.1 General Service List. The General Service List (GSL: [24]) is a curated list of 2000 highly frequent word families used in general English communication. It is typically broken into two lists, GSL1000, comprising the first 1000 most frequent words in the list, and GSL2000, comprising the second 1000 most frequent words in the list. For our own purposes, these two lists were each subdivided into five sublists (labeled A to E) on the basis of frequency (i.e., words in sublist A are more frequent than those in sublist B and so on).

2.3.2 Academic Word List. The Academic Word List (AWL: [6]) is a curated list of 570 highly frequent word families used in academic communication (e.g., journal articles, books, presentations). It is divided into 10 sublists, where the words in sublist 1 are more frequent than those in sublist 2, and so on.

3 METHODS

This section describes the methods used in the construction of the gold standard MCC question item set. In fact, two sets of items are in the process of being prepared: one based on GSL and another based on AWL.¹

3.1 Initial item set construction

For the GSL items, WQC was used to generate a total of 5,000 MCC items, with key words that were relatively evenly distributed across GSL1000 sublists A-E and GSL2000 sublists A-E. The corpus used to generate the stem sentences was the British Academic Written English corpus (BAWE: [7]). Although the target items were to test GSL words, a corpus of academic English was used due to the fact that we could not use an online corpus such as Wikipedia because the resulting items when used in online testing could easily be subject to a web search that would show the stem sentence source as the top hit. Hence, a corpus not indexed by search engines had to be used and WQC provided only BAWE as an option. The 5,000 GSL item list was reviewed to remove repetitive or obviously not well-formed items (e.g., sentence fragments extracted from a table), resulting in a list of 4,314 items.

For the AWL items, a total of 5,000 items had been generated earlier for use in our own training and testing program at the Center for English Language Education in Science and Engineering

(CELESE) at Waseda University in Tokyo, Japan. At that time, the list was evaluated by two teachers to remove obviously not well-formed or repetitive items as well as to simplify some stems (e.g., removing irrelevant parenthetical portions or adjunct phrases). This resulted in a set of 3,113 items.

3.2 Evaluation

The full set of GSL and AWL items (total 7,427 items) was then given to two experienced English as a foreign language (EFL) teachers in Japan for evaluation. Initially, they were both given two sublists: GSL1000 sublist B (386 items) and AWL sublist 10 (215 items).² Both checkers looked at all items in both of these sublists and were given the options below.

- Judge whether the item is usable (either as-is or with revision). Those that are unsalvageable were marked for removal from the dataset. These are referred to as 'rejected items'.
- Judge whether the item is well-formed as-is or not and mark it as such. Well-formed items were considered no further. These are referred to as 'well-formed items'.
- For each remaining usable item, revise it by light editing of the stem or changing one or more distractors as they thought suitable to make it well-formed. These are referred to as 'revised items'.

Items were displayed to the checkers in a spreadsheet (see Figure 2) in which they could mark the above options directly for each item and highlighting indicated their choices and revisions (discussed below). Rejected items were marked with a checkmark in the "Remove" column; Well-formed items were marked with *no* checkmark in the "Not well-formed" column; and revised items were marked with a checkmark in the "Not well-formed" column and then revisions were made to the stems and distractors in the respective "(revised)" columns.

The checkers were asked to judge well-formedness based on each item's suitability to test whether a university-level student of English in Japan knows enough about the key to distinguish it from the distractors. This was the practical goal of this task as we planned to begin using the final set of items for vocabulary training and testing in our own English for specific purposes (ESP) program for science and engineering students.

The items in (4)–(9) illustrate the decisions that checkers made. (4) was judged as a rejected item, likely because the stem sentence is too esoteric and in a form that is not readily salvageable. (5) was judged as a well-formed item as-is, likely because the trigram "the depressed economy" is highly salient compared to those with the distractors (e.g., "the conceived economy") and the stem sentence is not too long nor too difficult, containing several words semantically related to the key. (6) and (8) were judged as not well-formed, likely due to the overly obvious incorrectness of some distractors (in (6) and (8)) or excessively long stem (in (8)). These were revised to (7) and (9), respectively.

¹GSL (in particular) and AWL lists are regarded by some as "dated" and more modern versions of these lists with a similar aim have been produced (see e.g., Charles Browne's "New General Service List" [4] and "New Academic Word List" [3]). In the present work, we have continued to use GSL and AWL for legacy reasons, though we wish to eventually shift to newer lists in the future.

²The choice of these two sublists was a matter of convenience. We planned to use AWL sublist 10 immediately in CELESE classes on a pilot basis and wanted it completed first. As for GSL, the design of WQC caused very slow automatic generation of items for GSL1000 sublist A. Thus, we started the checker evaluation with GSL 1000 sublist B.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Checked	Key	Stem	Distractor 1	Distractor 2	Distractor 3	Not well-formed	Stem (revised)	Distractor 1 (revised)	Distractor 2 (revised)	Distractor 3 (revised)	Remove	
191	☒	straightforward	For cookies, it was very easy and _____ to find information in books or from the internet.	so-called	unconvinced	collapsible	☐	For cookies, it was very easy and _____ to find information in books or from the internet.	so-called	unconvinced	collapsible	☐	
192	☒	straightforward	Her concerns are _____ if their bodies are physically close together they are accompanied in death.	depressing	assembled	levy	☐	Her concerns are _____ if their bodies are physically close together they are accompanied in death.	depressing	assembled	levy	☒	
193	☒	straightforward	It is _____ for the addressee to understand the speaking setting.	persisting	unconvinced	adjacent	☐	It is _____ for the addressee to understand the speaking setting.	persisting	unconvinced	adjacent	☐	
194	☒	straightforward	Navigating through the Web pages is fairly _____, but could be improved.	adjacent	intrinsic	posed	☐	Navigating through the Web pages is fairly _____, but could be improved.	adjacent	intrinsic	posed	☐	
195	☒	straightforward	Such interactions make it exceedingly difficult to give a _____ answer to the question.	so-called	conceiving	adjacent	☒	Such interactions make it exceedingly difficult to give a _____ answer to the question.	depressed	conceiving	adjacent	☐	
196	☒	straightforward	This is a _____ proof using some basic linear algebra.	forthcoming	persisting	enormous	☒	This is a _____ proof using some basic linear algebra.	forthcoming	persisting	convinced	☐	
	☒	unconvinced	While these attacks boast some _____	so-called	intrinsic	colleague	☐	While these attacks boast some _____	so-called	intrinsic	colleague	☐	

Figure 2: Items presented in spreadsheet for checkers to evaluate. Items were highlighted automatically depending on checkers' evaluation or revisions.

- (4) The will is the world _____ as unified.
key: conceived distractors: depressing, convincing, collapsible
- (5) An alternative way to stimulate the _____ economy is loose monetary policy.
key: depressed distractors: conceived, compiled, reluctant
- (6) No major difficulties have been _____ to date.
key: encountered distractors: convinced, assemble, conceiving
- (7) No major difficulties have been _____ to date.
key: encountered distractors: convinced, assembled, conceived
- (8) It is a process which _____ a stage by stage approach whereby decision making must occur after each step.
key: adopts distractors: expense, husbands, settling
- (9) The process _____ a stage by stage approach.
key: adopts distractors: expenses, husbands, paints

For revision of items, various possibilities were suggested to the checkers.

- Shorten stems by removing irrelevant portions such as parentheticals or adjunct phrases.
- Remove or simplify technical terms or other rare phrases that may obscure the key's target context in the stem.
- Make distractors more uniform by ensuring they are of the same part-of-speech as the key and are otherwise syntactically consistent with the context (e.g., by number and case)
- Simplify stem sentences by revising unusual or awkward syntactic or discursive structures; especially those that may obscure the key's immediate context.

The initial portion of the evaluation task (focusing on GSL1000 sublist B and AWL sublist 10) was completed by both checkers and is therefore used below to compute inter-checker agreement. Cohen's κ was computed using `CohenKappa()` in the DescTools

library (v. 0.99.45) in the R statistical computing environment (v. 4.1.2).

After completing this initial portion, the remainder of the GSL and AWL sublist items was split up between the two checkers evenly (with no overlap) for continued evaluation. However, because of our own logistical limitations (i.e., time and budget), checkers were not asked to look at all of the items in each sublist. Rather a goal was set to find 125 suitable items³ (either well-formed or revised) with keys across as many word families as possible in each sublist. This dataset is used for the secondary analysis below. Note that because of this procedure, we cannot analyze the checkers' work based on accept/reject judgments because we cannot be sure if unselected items were rejected items, or simply passed over to avoid word family redundancy in keys.

4 RESULTS OF ITEM EVALUATION

This section of the paper gives some descriptive statistics regarding the checkers' work with the item dataset.

4.1 Inter-checker comparison

At the initial stage of evaluation, the two checkers looked at items targeting words from GSL1000 sublist B and AWL sublist 10. The following analysis is based on 386 items from GSL1000 sublist B and 214 items from AWL sublist 10, which both checkers evaluated fully.

Table 1 shows the number of rejected, well-formed, and revised items identified by each checker for the GSL1000 sublist B and AWL sublist 10 items. The results show a clear difference between the GSL and AWL lists, especially with respect to rejection: Compared to the GSL items, very few AWL items were rejected. Furthermore, there was some difference between the checkers with respect to the judgement of well-formedness versus revision. In particular, for AWL items, checker 2 was much more likely to revise the items than checker 1. A post-hoc review of the revisions shows that checker 2 was more likely than checker 1 to shorten stems with AWL items,

³The number 125 was selected by considering an estimation based on the initial pace of the checkers, the hourly rate of remuneration, the size of our budget, and the available time-frame.

Table 1: Number of rejected, well-formed, and revised items by each checker for GSL1000-B and AWL 10 items ($n_{GSL} = 386$, $n_{AWL} = 214$)

List Checker	GSL 1000-B		AWL 10	
	1	2	1	2
Rejected items	153	176	8	19
Well-formed items	17	4	154	3
Revised items	216	206	52	192

Table 2: Agreement counts (percentages in parentheses) between two checkers for item rejection ($n_{GSL} = 386$, $n_{AWL} = 214$) and for revision of non-rejected (i.e., well-formed or revised) items ($n_{GSL} = 175$, $n_{AWL} = 159$)

	Item rejection		Item revision	
	GSL (%)	AWL (%)	GSL (%)	AWL (%)
Agree	293 (75.9)	191 (89.3)	159 (90.9)	52 (27.5)
Disagree	93 (24.1)	23 (10.7)	16 (9.1)	137 (72.5)
Cohen's κ	0.51	0.10	0.17	0.01

Table 3: Agreement counts (percentages and Cohen's κ in parentheses) between two checkers for revisions in revised items ($n_{GSL} = 157$, $n_{AWL} = 49$). Note that rows are non-exclusive and thus the percentages do not add up to 100%.

Revision	GSL (% , κ)	AWL (% , κ)
Stem	101 (64.3, 0.31)	12 (24.5, 0.00)
Distractors	85 (54.1, 0.47)	15 (30.6, 0.29)
Stem and distractors	57 (36.3, 0.40)	5 (10.2, 0.10)

shortening them by an average 5.1 characters more than checker 1 [$t(48) = 2.98$, $p < .005$].

Table 2 shows the number of items for which the checkers agreed on their identification as rejected or non-rejected items. On the question of item rejection, the results show agreement at a rate of 75.9% ($\kappa = 0.51$) for GSL items and 89.3% ($\kappa = 0.10$) for AWL items—a combined rate of 80.7% ($\kappa = 0.54$). Table 2 also shows the number of non-rejected items (i.e., well-formed or revised items) which the checkers agreed on their need for revision or not. The results show agreement for GSL items at 90.9% (but low $\kappa = 0.17$, likely because checkers chose to revise 98.7% of items), but a lower level of agreement for AWL items at 27.5% ($\kappa = 0.01$), yielding a combined rate of 58.0% ($\kappa = 0.02$).

Table 3 focuses on only revised items (i.e., those items which both checkers revised) and shows the number of items for which the checkers agreed on what sort of revision to undertake: revising the stem, revising distractors, and revising both the stem and distractors (note that in the table, these categories overlap and thus percentages do not add up to 100%). The combined agreement rate for revising the stems was 54.8% ($\kappa = 0.22$), for distractors was 48.5% ($\kappa = 0.44$), and for both stem and distractors was 30.1% ($\kappa = 0.34$). The results show that the agreement rate on GSL items was much higher

Table 4: Number of items judged as well-formed or revised items by each checker for all GSL and AWL sublist items ($n_{GSL} = 1453$, $n_{AWL} = 1339$)

List Checker	GSL		AWL	
	1	2	1	2
Well-formed items	43	4	205	6
Revised items	847	559	571	557
Total	890	563	776	563

Table 5: Counts (percentages in parentheses) of types of revisions applied to revised items in GSL and AWL items ($n_{GSL} = 1406$, $n_{AWL} = 1128$)

	GSL (%)	AWL (%)
Stem	1149 (81.7)	734 (65.1)
1 distractor	229 (16.3)	297 (26.3)
2 distractors	396 (28.2)	351 (31.1)
3 distractors	648 (46.1)	360 (31.9)

than that for AWL items. Though not described here, the actual revisions undertaken were different between the checkers: Each changed the stem in different ways to make it suitable and changed the distractors to different alternatives in order to make the item suitable in their judgment. A detailed description and analysis of these changes is beyond the scope of the present paper.

4.2 Status of full item dataset

In this section, the results for the checkers' work on all the GSL and AWL sublist items are described. As explained above at the end of Section 3.2, this analysis excludes rejected items. The total number analyzed here is 2,792 items.

Table 4 shows the number of items judged by each checker as well-formed or revised items from both the GSL and AWL sublists. Overall, they judged 9.2% of items as well-formed as-is and 90.8% of items as needing revision. The results show that both checkers leaned strongly toward judging items as in need of revision, though checker 1 was somewhat more lenient toward the AWL items than was checker 2. This is consistent with the inter-checker observation above showing that the agreement for AWL revision judgment was lower than that for GSL.

Table 5 shows the number of times checkers made various revisions to the revised items in the GSL and AWL item sets (with overlap between stem and distractor revision counts). Overall, checkers revised 74.3% of the stems and 66.4% of the distractors. The results show that checkers were more likely to revise the GSL stems than the AWL stems ($\chi^2(1) = 90.0$, $p < .001$). In fact, the size of the change also differed: checkers shortened the GSL stems by 19.6 characters but the AWL stems by only 9.6 characters [$t(2345) = 11.6$, $p < .001$]. The results further show that checkers tended to revise more of the GSL distractors, revising all 3 distractors in almost half of cases (46.1%, $\chi^2(3) = 64.7$, $p < .001$).

5 DISCUSSION

The dataset at the focus of this study is a collection of MCC question items for use in vocabulary training and testing for learners of English as a second/foreign language. This set of data was evaluated by two checkers for well-formedness and each item was either rejected completely as unsalvageable or retained by either accepting it as-is or revising it in some manner to make it suitable for use. Analysis of a portion of the dataset which both checkers reviewed showed a high inter-checker agreement for rejection (80.7%, $\kappa = 0.54$), though a lower agreement for judgment of the need for revision (58.0%, $\kappa = 0.02$). Following the scheme in Landis and Koch [12] for interpreting Cohen's κ , the strength of these agreement levels is "moderate" and "slight", respectively. However, the revision agreement rates for GSL items were higher than those for the AWL items, almost reaching a "fair" strength of agreement. For the whole dataset, the checkers were more likely to revise the GSL items than the AWL items, and to make more revisions to each item (reduce stem length more; revise more distractors) for the GSL items than the AWL items.

At least part of the difference between the GSL and AWL items is likely due to the provenance of these two subsets, as explained in Section 3.1: The AWL items came from a collection of items that had already been reviewed and revised for use in online testing in our program at Waseda University. Thus, these items were more likely to have been seen as well-formed as-is, or in need of only small revisions for improved suitability. The much lower agreement level for types of revisions in the AWL than the GSL sets is further explained by the fact that many more items (in fact, nearly all: 98.7%) in the GSL set were judged in need of revision: Thus, it is more likely that the checkers' judgements would coincide (cf., a "ceiling effect"), while the lower perceived necessity of revision in the AWL items would mean that the checkers' judgements could more easily disperse across the items.

6 FUTURE PLANS

The combined set of well-formed and revised items (total 2,792) has already been used with learners in a pilot experiment (described in [20]) using a simple app platform to provide training and testing of learners' vocabulary knowledge. In the training phase of this app, one feature is that after a user is initially presented with a MCC question, as time passes, distractors gradually disappear, giving them a hint toward the correct answer. In this way, the learner's response latency is tied to nudges to help them discover the right answer. Although not implemented yet, we see many further ways to use learners' response latency to improve the learning experience. For example, the latency can be recorded and used in an adaptive testing approach (cf., [23]) to inform how soon a particular target word for learning is recycled in the training phase. That is, greater latency would promote a shorter cycle to the next repetition of the target word.

Latency information could also be used as a means for evaluating the efficacy of the distractors. If distractors are equally effective distractors, one would expect that after each one disappears, learners might show uniform behavior in deciding which of the remaining options is the correct answer. However, if, for example, one particular distractor's disappearance never leads to quick(er) recognition

of the correct answer, that suggests the distractor is not really doing any work as a distractor: That is, its disappearance is not helping the learner (i.e., from Shannon's [21] information theoretic point of view, the distractor has low information value or informativity).

Future plans with the MCC item dataset include asking checkers to complete an evaluation of the complete set of over 7,000 items total. This set will then be used together with an existing natural language generation tool (e.g., BERT, GPT-3) to generate novel MCC questions focusing on a specified target word. These items can then be used in training or testing second language learners' vocabulary knowledge using the app described above. In that context, sensing learners' response latency for distractors can serve as a means to evaluate the validity and efficacy of the generation of novel items.

More sophisticated sensing technologies such as eye-tracking might also be used with learners while they are engaged in practice sessions to gauge difficulty levels or efficacy of stem sentences and distractors. Slow reading times over stem sentences may suggest they are too difficult for the key word that is targeted in the item. Similarly, fewer fixations on certain distractors may suggest that they are not effectively serving as distractors relative to other distractors and the key word.

Researchers who may be interested in examining the inter-checker information contained in the dataset described in this paper are invited to contact the authors to inquire about how they may access and use the dataset in their own research and projects.

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